METRIC

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13 June 2007

SUPERSEDING
MIL–M–11361D
24 December 1992

DETAIL SPECIFICATION

MAGNESIUM CARBONATE

Inactive for new design after 14 Oct 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers magnesium carbonate to be used in smoke mixtures.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 6810

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FEDERAL STANDARDS

FED−STD−313 − Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ENVIRONMENTAL PROTECTION AGENCY

EPA Method 6010B − Inductively Coupled Plasma−Atomic Emission Spectrometry

(Copies of these documents are available online at www.epa.gov or from the U.S. Environmental Protection Agency, Ariel Rios Building, 1200 Pennsylvania Avenue, N.W., Washington, DC 20460.)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

B 527 − Standard Test Method for Determination of Tap Density of Metallic Powders and Compounds

(Copies of this document are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428−2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Chemical and physical characteristics. The magnesium carbonate shall conform to the chemical and physical characteristics of table I.

TABLE I. Chemical and physical requirements.

Characteristic	Requirement	Test Paragraph
Assay (as MgO), percent by weight	40.0 to 43.5	4.4.1
Matter insoluable in HCl, percent by weight, maximum	0.02	4.4.2
Water−soluable substances, percent by weight, maximum	0.50	4.4.3
Calcium (Ca), percent by weight, maximum	0.45	4.4.4
Arsenic (As), ppm, maximum	2	4.4.5
Lead (Pb), ppm, maximum	10	4.4.5
Other Heavy metals, ppm maximum	20	4.4.5
Iron (Fe), percent by weight, maximum	0.02	4.4.6
Moisture, percent by weight, maximum	1.0	4.4.7
Particle size distribution:		4.4.8
Average particle size, microns, maximum	20	
Average particle size, microns, minimum	7	
Distribution shape	Single mode	
Tap density, grams per milliliter, minimum (dry basis)	0.2	4.4.9
Tap density, grams per milliliter, maximum (dry basis)	0.6	4.4.9
Thermal Characteristics:		4.4.10
Peak of first endotherm, temp. range (°C)	235 to 250	
Loss after first endotherm, percent by weight, minimum	4	
Peak of second endotherm, temp. range (°C)	435 to 450	

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

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4.2 First article inspection.

4.2.1 Lotting. The first article lot shall consist of at least 50 kg of magnesium carbonate manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract. If there is any change of vendors, methods, materials, equipment, facilities, or if there is a break in production of 90 days or more, the contracting Officer shall be notified and at the discretion of the Government, a new first article sample may be required.

4.2.2 Sampling. The contractor shall randomly select three representative samples, of approximately one kg each, from the first article lot of magnesium carbonate and place each sample in a clean dry container labeled to identify the first article lot.

4.2.3 Inspection procedure. Each first article sample shall be subjected to all of the inspections specified in the classification of characteristics in 4.3.5.

4.2.4 Acceptance criteria. If any first article sample fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the magnesium carbonate produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, and at the same time. The minimum lot size shall be 500 kg. In the event the lot is a batch operation (see 6.7), several batches may be mixed together to form a lot, provided the batches are subjected to some physical mixing operation intended to make the final product uniform and homogenous.

4.3.2 Sampling. The contractor shall randomly select three representative samples, of approximately one kg each, from the lot of magnesium carbonate and place each sample in a clean dry container labeled to identify the lot represented.

4.3.3 Inspection procedure. Each magnesium carbonate sample shall be inspected in accordance with the classification of characteristics in 4.3.5. Failure of any sample to conform to any characteristic in the classification of characteristics based acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals who use or maintain the product. Characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item are also critical characteristics. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the

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item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance inspections shall be as specified in the following classification of characteristics paragraphs.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.3.5(a)	Magnesium carbonate	SHEET 1 OF 1	

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Chemical and physical properties	Note 1.	3.2	
	a. Assay			4.4.1
	b. Matter insoluable in HCl			4.4.2
	c. Water soluable matter			4.4.3
	d. Calcium content			4.4.4
	e. Arsenic, lead and other heavy metals content			4.4.5
	f. Iron content			4.4.6
	g. Moisture content			4.4.7
	h. Particle size			4.4.8
	i. Tap density			4.4.9
	j. Thermal characteristics			4.4.10

NOTES:

1. Sampling shall be in accordance with 4.3.2. One specimen shall be taken from each sample to perform each test.

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4.4 Tests.

4.4.1 Assay (see 6.5). Dissolve 5 g of 8−hydroxyquinoline (oxine) in 12 milliliters (mL) of glacial acetic acid and dilute to 100 mL with water. Filter if necessary. Weigh to the nearest 0.1 milligram (mg) approximately 0.2 g of specimen and transfer to a 250 mL beaker. Add 30 mL of water and just sufficient concentrated hydrochloric acid to dissolve the specimen. Dilute to about 100 mL with water. Dry a filtering crucible to constant weight at 105 to 110°C, cool in a desiccator, and weigh to the nearest 0.1 mg. Add 2 g of ammonium chloride to the specimen solution and add concentrated ammonium hydroxide to a pH of 10 to 11. Heat the solution to 70 to 80°C and add dropwise, while stirring, 13 mL of the 8−hydroxyquinoline solution. Add more ammonium hydroxide, if necessary, to maintain the pH at 10 to 11. While keeping the temperature between 70 to 80°C, mechanically stir the mixture for 15 minutes. Filter through the weighed crucible. Wash the precipitate well with small portions of a 1 percent ammonium hydroxide solution. Dry and ignite the precipitate at 800°C for 20−30 minutes, cool in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent magnesium oxide as follows:

$$\text{Percent magnesium oxide} = \frac{100A}{W}$$

where: A = grams of precipitate, and
W= grams of specimen

4.4.2 Matter insoluble in hydrochloric acid. Dry a filtering crucible at 100 to 105°C, cool in a desiccator, and weigh to the nearest 0.1 mg. Transfer approximately 10 g of the specimen, weighed to the nearest 0.01 g, to a 250 mL beaker, mix with 100 mL of water, and add slowly, while stirring, 25 mL of concentrated hydrochloric acid. Boil the solution for 5 minutes and filter while hot. Wash the residue thoroughly with hot water and dry to constant weight at 100 to 105°C. Cool in desiccator and weigh to the nearest 0.1 mg. Calculate the percent matter insoluble in hydrochloric acid as follows:

$$\text{Percent matter insoluble in hydrochloric acid} = \frac{100A}{W}$$

where: A = grams of residue, and
W= grams of specimen

4.4.3 Water−soluble substances. Weigh to the nearest milligram approximately 2 g of the specimen and boil with 50 mL of water for 5 minutes. Restore the original volume with hot water and filter while hot. Transfer 25 mL of the clear filtrate to a weighed platinum dish and place on a steam bath to evaporate the water. Evaporate to dryness at 105 to 110°C. Cool in a desiccator and weigh rapidly. Calculate the percent water−soluble substances as follows:

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$$\text{Percent water−soluble substances} = \frac{100A}{W}$$

where: A = grams of residue, and
W = grams of specimen in aliquot used

4.4.4 Calcium content (see 6.5).

4.4.4.1 Standard magnesium solution. Weigh to the nearest 0.1 mg approximately 0.5 g of polished magnesium metal ribbon and dissolve in 50 mL of hydrochloric acid (1 to 10).

4.4.4.2 Ammonium hydroxide−ammonium chloride buffer. Dissolve 60 g of ammonium chloride in about 200 mL of water, add 570 mL of concentrated ammonium hydroxide, and dilute to 1 liter.

4.4.4.3 Eriochrome black T indicator solution. Dissolve 0.15 g of Eriochrome black T and 0.50 g of borax in 35 mL of methanol. Warm to effect solution.

4.4.4.4 Standard disodium dihydrogen ethylene diaminetetraacetate solution. Dissolve 8.00 g of disodium dihydrogen EDTA in 2 liters of water. Pipet 10 mL of the standard magnesium solution into a 250 mL Erlenmeyer flask. Add 25 mL of water, 10 mL of the ammonium hydroxide−ammonium chloride buffer, and 3 or 4 drops of the indicator solution. Titrate against a white background under an incandescent light with the disodium dihydrogen EDTA solution to a pure end point having no trace of pink. Calculate the molarity of the disodium dihydrogen EDTA solution as follows:

$$M = \frac{0.8224A}{V}$$

where: M = molarity of the disodium dihydrogen EDTA solution,
A = grams of magnesium ribbon used in the preparation of the standard magnesium solution, and
V = milliliters of disodium dihydrogen EDTA solution used.

4.4.4.5 Standard calcium solution. Dissolve 1.0009 g of calcium carbonate in 5 mL of hydrochloric acid (1 to 1) and dilute to 1 liter with water. Pipet 30 mL of this solution into a 250 mL Erlenmeyer flask, add 25 mL of water, 2 mL of 20 percent sodium or potassium hydroxide solution, and 1 tablet of murexide (ammonium purpurate) indicator (see 6.8). Titrate with the standard disodium dihydrogen EDTA solution against a white background to match a blank prepared in the same manner. Calculate the molarity of the calcium solution as follows:

$$\text{Molarity of calcium solution} = \frac{MV}{30}$$

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where: M = molarity of the disodium dihydrogen EDTA solution, and
V = milliliters of disodium dihydrogen EDTA solution used

4.4.4.6 Procedure. Weigh to the nearest milligram approximately 2 g of the specimen and transfer to a 250 mL Erlenmeyer flask. Add 10 mL of water and 5 mL of concentrated hydrochloric acid in small portions, being careful not to lose any of the specimen by spattering. When solution is complete, dilute to about 25 mL. Filter the solution through medium−texture filter paper into a 250 mL beaker and wash the filter paper and residue with small potions of hot water, not exceeding a total of 50 mL. Carefully evaporate the filtrate to dryness. Dissolve the residue in 9 mL of water, add 1 mL of sulfuric acid (1 to 4) and cool. Slowly add 90 mL of methanol, stirring constantly. Allow the solution to stand for about 2 hours until the precipitation is complete. Filter through a porcelain filtering crucible. Transfer the last traces of precipitate to the filtering crucible by means of a rubber policeman and wash with small portions of methanol (9 to 1). Dissolve the residue in 25 mL of cold water. Adjust the pH of the solution to 10, using 20 percent sodium or potassium hydroxide, and add 1 tablet of murexide indicator (see 6.8). Prepare a blank consisting of 25 mL of water adjusted to a pH of 10 and add 1 murexide tablet. Titrate the specimen solution with standard disodium dihydrogen EDTA solution against a white background to match the color of the blank. The color change is from pink to violet and occurs gradually. If there is doubt about the end point, the solution may be back−titrated with the standard calcium solution. Calculate the percent calcium in the sample as follows:

$$\text{Percent calcium} = 4.008 \times \frac{(VN) - (BM)}{W}$$

where: V = milliliters of standard disodium dihydrogen EDTA solution used
N = molarity of disodium dihydrogen EDTA solution,
B = milliliters of standard calcium solution used,
M = molarity of calcium solution, and
W = grams of specimen used

4.4.5 Arsenic, lead and other heavy metal contamination. Determine the heavy metals content of the sample in accordance with EPA Method 6010B.

4.4.6 Iron content (see 6.5).

4.4.6.1 Ammonium thiocyanate solution. Dissolve 25 g of ammonium thiocyanate in water and dilute to 100 mL.

4.4.6.2 Standard iron solution. Dissolve 0.8635 g of ferric ammonium sulfate ($FeNH_4(SO_4)_2 - 12\ H_2O$) in 10 mL of sulfuric acid (5 to 95) and dilute to exactly 1 liter. Dilute 10 mL of this solution to 100 mL with water.

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4.4.6.3 Procedure. Dissolve 0.5 g of the specimen in 10 mL of hydrochloric acid (1 to 3) and boil gently for 1 minute. Cool and dilute to 50 mL. Add 30 to 50 mg of ammonium persulfate crystals and 3 mL of the ammonium thiocyanate solution. Prepare a control sample containing 1 mL of standard iron solution, 4 mL of hydrochloric acid (1 to 1), 45 mL of water, and the same quantity of ammonium persulfate and ammonium thiocyanate used above. Compare the specimen and standard in matched color comparison tubes. If the color produced by the specimen is darker than that of the control sample, the specimen contains more than 0.002 percent iron.

4.4.7 Moisture. Transfer approximately 0.2 g of the specimen, weighed to the nearest 0.1 mg, to a dry, tared weighing bottle. Desiccate for 3 hours under an absolute pressure of 24 mm of mercury at 25°C ± 5°C. Reweigh the bottle and specimen. Calculate the loss in weight as percent moisture.

4.4.8 Particle size. Particle size analysis shall be performed using the laser diffraction method outlined below. Particle size analysis and histogram are required with material certification.

The sample size required for particle size analysis should follow the recommendation of the specific manufacturer of the instrument being used. Typical sample sizes are a few milligrams and should be taken to ensure representative sampling of the magnesium carbonate has occurred. The particle size analysis is to be performed in deionized water with a few drops of surfactant added to the sample/water prior to analysis. Internal sample sonication should occur for 60 seconds at 20 to 30 watts power setting prior to use for analysis. External sonication using an ultrasonic probe may be used if the equipment is not equipped with such capability.

Prior to analysis, sonicated sample should be loaded in the instrument to enable sufficient obscuration of the laser diffraction signal, typically 5−35% to ensure optimal laser diffraction signal to noise ratio. The sample should be run for a minimum of 60 seconds at a flow rate that does not introduce air bubbles into the sample chamber. A minimum of three runs must be performed.

Data analysis shall use a refractive index of 1.51 for the magnesium carbonate and 1.33 for the deionized water. The calculations for particle size should be based on Mie scattering theory. The mean particle size should be reported as an average of three runs. The 10th, 50th, and 90th percentile of particle size and a histogram of the particle distribution shall also be reported as subjective information.

4.4.9 Tap density. Determine the tap density of the sample in accordance with ASTM B527 with the following parameters:

Test cylinder capacity = 100 cm^3
Weight of sample = 50 g

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4.4.10 Thermal characteristics. This method is used to determine the weight loss and reaction (peak) temperatures of magnesium carbonate as it is heated from ambient temperature to the specified temperature. The test is conducted at a controlled heating rate in a controlled environment. The magnesium carbonate shall be run on either a Differential Scanning Calorimeter (DSC) or a Differential Thermal Analyzer (DTA) to determine reaction temperature. The weight loss of the magnesium carbonate sample shall be determined on a Thermogravimetric Analyzer (TGA). A Simultaneous Differential Scanning Calorimeter/Thermogravimetric Analyzer (SDT) may also be used to collect reaction temperature and weight loss concurrently.

The thermal analysis shall be performed in a dry, inert gas environment (99% pure argon) at a flow−rate of 25−mL/min to 100−mL/min. The flow rate is specific to each instrument. The sample shall be run from ambient to 550˚C at a rate of 10˚C/min + 0.1˚C/min. Typical sample weight is three to seven milligrams. Larger sample size may be use based on the instrument manufacturer's recommendation.

The heat flow (DSC), temperature difference (DTA) and weight loss (TGA) shall be collected and plotted versus temperature. A temperature transition is marked with a peak when the sample absorbs (endothermic) or releases (exothermic) energy. The peak temperatures shall be determined from the derivatives of the DSC or DTA plot. The weight loss shall be determined directly from the TGA plot. The analysis shall be conducted on two samples from the same lot and reported as an average. The reported data shall include temperature of first peak, temperature of second peak, and weight loss of the sample from ambient temperature to 300˚C.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Magnesium carbonate, covered by this specification is intended for use in the manufacture of signaling smoke mixtures.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article:
(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.
(c) Packaging requirements (see 5.1).
(d) Material Safety Data sheets preparation (see 6.4)
(e) Certificate of Analysis (see 6.5)

6.3 Subject term (key word) listing.
calcium (Ca)
hydrochloric acid (HCl)
iron (Fe)
lead (Pb)
magnesium oxide (MgO)

6.4 Material Safety Data Sheets. Material Safety Data Sheets for the magnesium carbonate should be prepared and submitted in accordance with Fed. Std. No. 313.

6.5 Assay, calcium, and iron. When approved by the contracting officer, the contractor may use an atomic absorption procedure.

6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Preparing activity:
Army – EA
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Army – MD, SM

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